

HW1

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CPSC 321 — Gonzaga University
compiled with L^AT_EX and TikZ :)

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part A

question 1

```
AIRPORT(air_id, air_name, city, state)
FK: none
```

`air_id` is the primary key since its unique, and minimal.

air_id	air_name	city	state
GEG	Spokane International	Spokane	WA
SEA	Seattle-Tacoma International	Seattle	WA
LAX	Los Angeles International	Los Angeles	CA

question 2

```
AIRLINE(air_code, airline_name, air_id, founded_year)
FK: air_id -> AIRPORT(air_id)
```

`air_code` is the primary key since its unique, and minimal. `air_id` is a foreign key referencing the hub airport's `air_id` in AIRPORT.

air_code	airline_name	air_id	founded_year
AS	Alaska Airlines	SEA	1932
WN	Southwest Airlines	LAX	1967
UA	United Airlines	GEG	1926

question 3

```
FLIGHT_ROUTE(air_code, flight_num, dep_id, arr_id, distance)
FK: air_code -> AIRLINE(air_code)
FK: dep_id -> AIRPORT(arr_id)
FK: arr_id -> AIRPORT(arr_id)
```

primary key is composite. neither attribute is unique on its own: an airline operates many routes, so `air_code` repeats, and two airlines can both use flight number 122, so `flight_num` repeats. together they are unique, and removing either one breaks uniqueness, so the pair is minimal.

`dep_id` and `arr_id` are both foreign keys into `AIRPORT`.

<u>air_code</u>	<u>flight_num</u>	<u>dep_id</u>	<u>arr_id</u>	<u>distance</u>
AS	122	GEG	SEA	224
AS	345	SEA	LAX	954
WN	122	LAX	GEG	1046

question 4

```
PASSENGER(pass_id, first_name, last_name, email)
Additional candidate key: {email}
FK: none
```

`pass_id` and `email` are both candidate keys since both are stated to be unique. `pass_id` should be the primary key because email addresses can change, and that would be bad.

<u>pass_id</u>	<u>first_name</u>	<u>last_name</u>	<u>email</u>
1	Jack	Smith	jsmith@example.com
2	Alan	Cho	acho@example.com
3	Greyson	Knippel	gknippel@example.com

```
BOOKING(pass_id, air_code, flight_num, travel_date, seat, price)
FK: pass_id -> PASSENGER(pass_id)
FK: (air_code, flight_num) -> FLIGHT_ROUTE(air_code, flight_num)
```

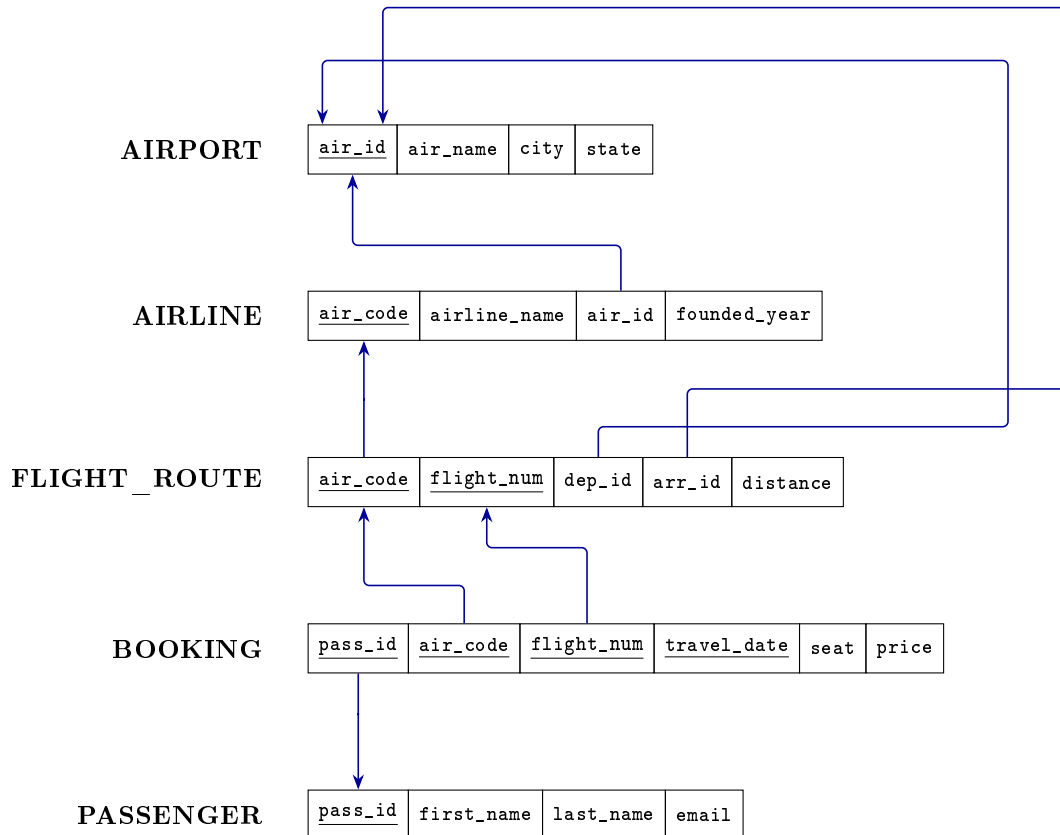
the constraint “booked once per route per date” makes those four attributes the key. every removal of the attributes breaks it as well.

(`air_code`, `flight_num`) is a composite foreign key the parent key is composite, so the reference has to carry both columns together. `seat` and `price` are irrelevant to this table’s keys.

<u>pass_id</u>	<u>air_code</u>	<u>flight_num</u>	<u>travel_date</u>	seat	price
1	AS	122	2026-10-02	14A	189.50
1	AS	122	2026-11-15	08C	210.00
2	AS	122	2026-10-02	14B	205.75

part B

question 5



the underlined attributes are primary keys. each arrow runs from a foreign key to the primary key it references. the two arrows from BOOKING into FLIGHT_ROUTE are a single composite foreign key: air_code and flight_num must be carried together, since the key they reference is composite. AIRPORT is referenced three times: once by AIRLINE for its hub, and twice by FLIGHT_ROUTE for the departure and arrival airports.

part C

question 6

(a)

```
MaintenanceRecord(aircraft_id, maintenance_date, mechanic_id,  
    aircraft_model, manufacturer, mechanic_name, work_performed,  
    parts_cost, mechanic_cert_level)  
FK: none
```

the key means one row is “this mechanic worked on this aircraft on this date,” so other attributes should relate to the situation.

the redundant values are `mechanic_name` and `mechanic_cert_level`. both are facts about the mechanic, not about a maintenance job, so they repeat on every record that mechanic performs:

attribute	really a fact about	so it repeats
<code>aircraft_model</code>	the aircraft	every record for that aircraft
<code>manufacturer</code>	the aircraft (and its model)	every record for that aircraft
<code>mechanic_name</code>	the mechanic	every record by that mechanic
<code>mechanic_cert_level</code>	the mechanic	every record by that mechanic
<code>work_performed</code>	the visit itself	never
<code>parts_cost</code>	the visit itself	never

a 737-900 is always built by Boeing, so the same value also repeats across completely different aircraft that happen to share a model.

only `work_performed` and `parts_cost` change from one maintenance visit to the next, so they are the only attributes that actually belong in this table.

aircraft	date	mech	model	manuf.	mech_name	work	cost	cert
N101AS	2026-03-04	M7	737-900	Boeing	James Smith	brake pads	1450.00	A&P
N101AS	2026-06-18	M7	737-900	Boeing	James Smith	hydraulics	0.00	A&P
N220WN	2026-06-18	M7	737-700	Boeing	James Smith	oil change	310.00	A&P
N101AS	2026-09-02	M3	737-900	Boeing	James Smith	tires	2100.00	IA

“James Smith” and “A&P” are written three times because M7 appears three times, and “Boeing” is written on all four rows despite two different aircraft and two different models.

(b)

each group of facts from part (a) moves into its own table, and the keys are whatever decides it. what is left behind in the original table is just the id, which links back to the new table:

```
AIRCRAFT_MODEL(aircraft_model, manufacturer)
FK: none

AIRCRAFT(aircraft_id, aircraft_model)
FK: aircraft_model -> AIRCRAFT_MODEL(aircraft_model)

MECHANIC(mechanic_id, mechanic_name, mechanic_cert_level)
FK: none

MAINTENANCE_RECORD(aircraft_id, maintenance_date, mechanic_id,
    work_performed, parts_cost)
FK: aircraft_id -> AIRCRAFT(aircraft_id)
FK: mechanic_id -> MECHANIC(mechanic_id)
```

the key of MAINTENANCE_RECORD does not change: it is still the same three attributes but `aircraft_id` and `mechanic_id` are now foreign keys as well as key attributes. only `work_performed` and `parts_cost` stay behind, since those are the only values that actually describe the visit.

the fourth table is what removes the last of the redundancy. if `manufacturer` were left in AIRCRAFT, “Boeing” would still have to be written once per aircraft. giving the model its own table means each manufacturer is recorded exactly once.

<u>aircraft_model</u>	manufacturer	<u>aircraft_id</u>	<u>aircraft_model</u>
737-900	Boeing	N101AS	737-900
737-700	Boeing	N220WN	737-700
A320-200	Airbus	N455UA	A320-200

<u>mechanic_id</u>	<u>mechanic_name</u>	<u>mechanic_cert_level</u>
M7	Dana Weiss	A&P
M3	Omar Diaz	IA
M9	Lena Petrov	A&P

<u>aircraft_id</u>	<u>maintenance_date</u>	<u>mechanic_id</u>	<u>work_performed</u>	<u>parts_cost</u>
N101AS	2026-03-04	M7	brake pads	1450.00
N101AS	2026-06-18	M7	hydraulics	0.00
N220WN	2026-06-18	M7	oil change	310.00
N101AS	2026-09-02	M3	tires	2100.00

these have the same information as the original table but are just more optimized.